summary statistics for BYD returns

The MEANS Procedure

Analysis Variable : ret				
N	Mean	Std Dev	Minimum	Maximum
500	1.0782942	1.1850245	-2.7685660	7.0088740

BYD returns

The UNIVARIATE Procedure
Variable: ret

Moments			
N	500	Sum Weights	500
Mean	1.07829422	Sum Observations	539.14711
Std Deviation	1.18502452	Variance	1.40428312
Skewness	0.40237677	Kurtosis	1.49698349
Uncorrected SS	1282.09649	Corrected SS	700.737278
Coeff Variation	109.898069	Std Error Mean	0.05299591

Basic Statistical Measures			
Location		Variability	
Mean	1.078294	Std Deviation	1.18502
Median	1.029292	Variance	1.40428
Mode	.	Range	9.77744
		Interquartile Range	1.39302

Tests for Location: Mu0=0				
Test	Statistic		p Value	
Student's t	t	20.34674	Pr > |t|	<.0001
Sign	M	168.5	Pr >= |M|	<.0001
Signed Rank	S	51471	Pr >= |S|	<.0001

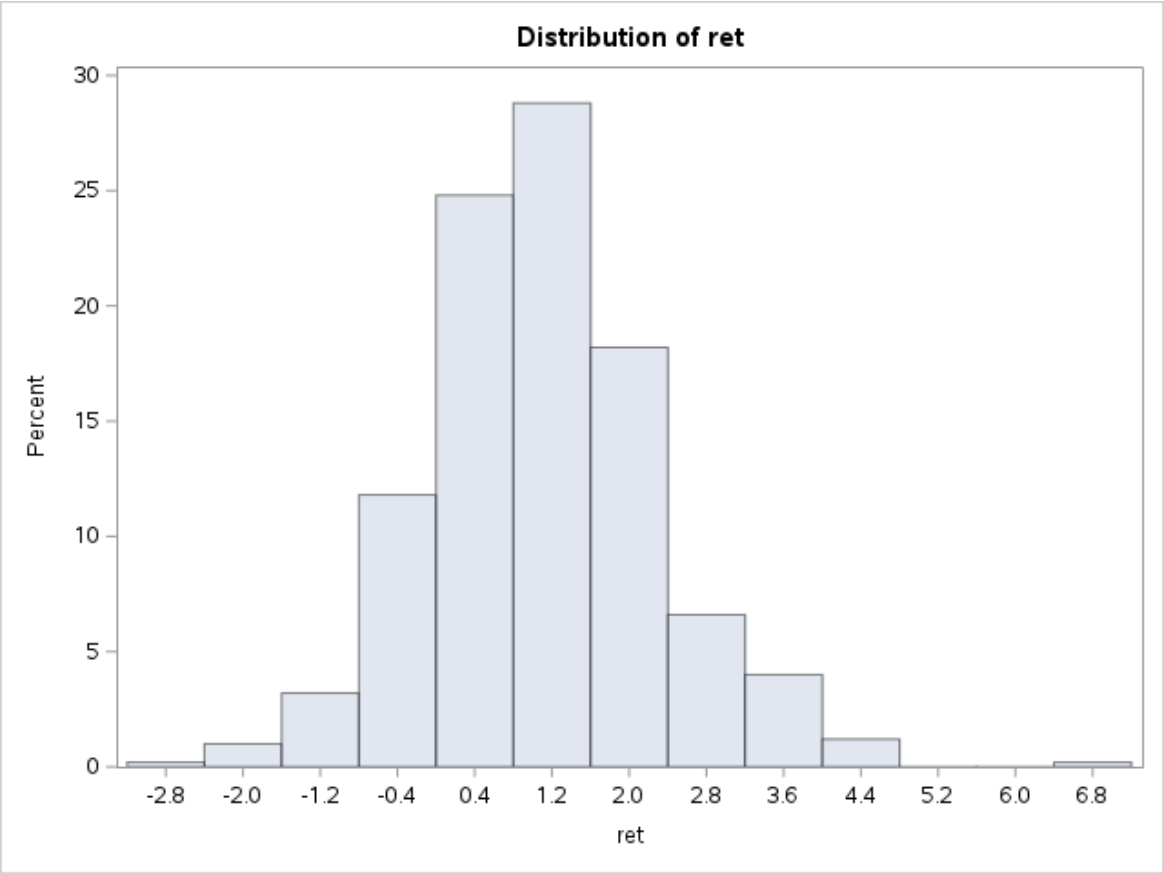
Tests for Normality				
Test	Statistic		p Value	
Shapiro-Wilk	W	0.984533	Pr < W	<0.0001
Kolmogorov-Smirnov	D	0.047025	Pr > D	<0.0100
Cramer-von Mises	W-Sq	0.271357	Pr > W-Sq	<0.0050
Anderson-Darling	A-Sq	1.633679	Pr > A-Sq	<0.0050

Quantiles (Definition 5)	
Level	Quantile
100% Max	7.008874
99%	4.232911
95%	3.261007
90%	2.558422
75% Q3	1.748441
50% Median	1.029292
25% Q1	0.355425
10%	-0.364024
5%	-0.684184
1%	-1.717957
0% Min	-2.768566

Extreme Observations			
Lowest		Highest	
Value	Obs	Value	Obs
-2.76857	448	4.37690	381
-2.29995	364	4.47216	348
-2.11206	196	4.57039	399
-1.79588	368	4.76663	450
-1.77583	446	7.00887	369

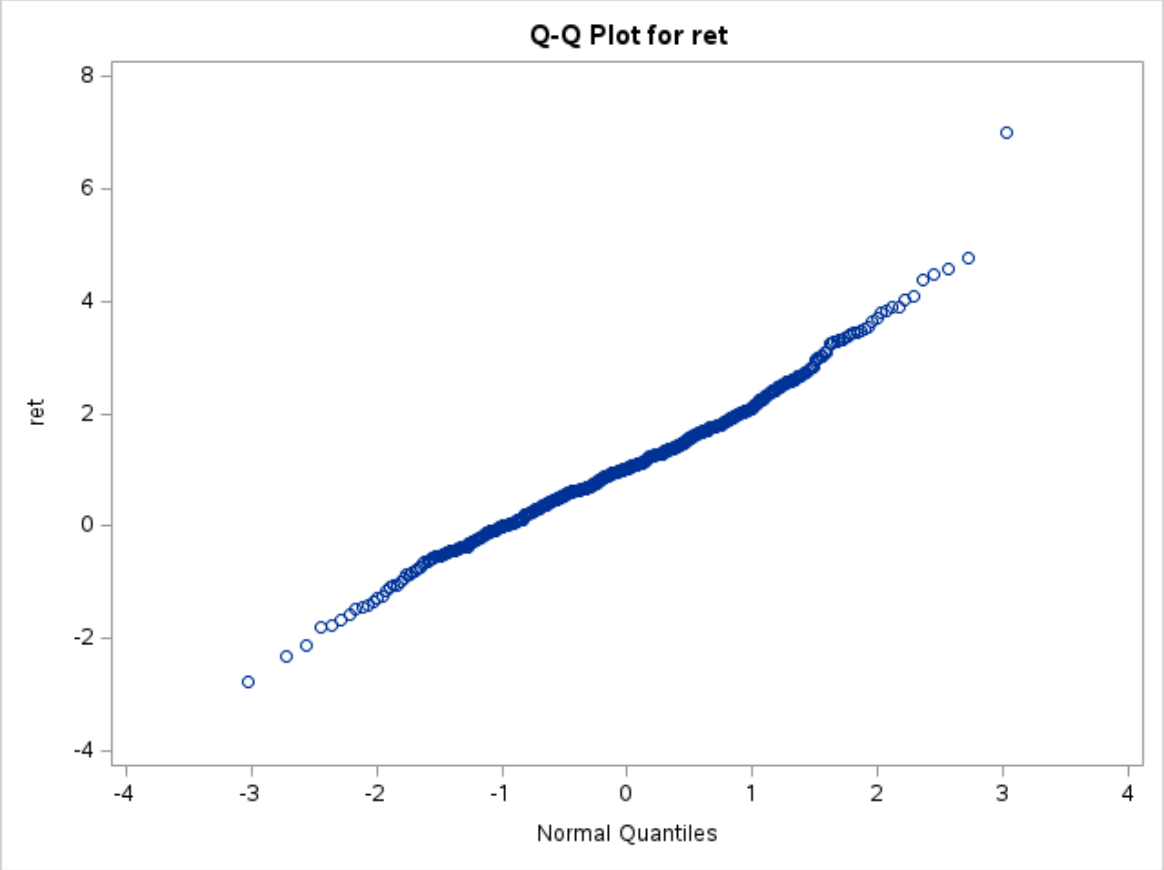
BYD returns

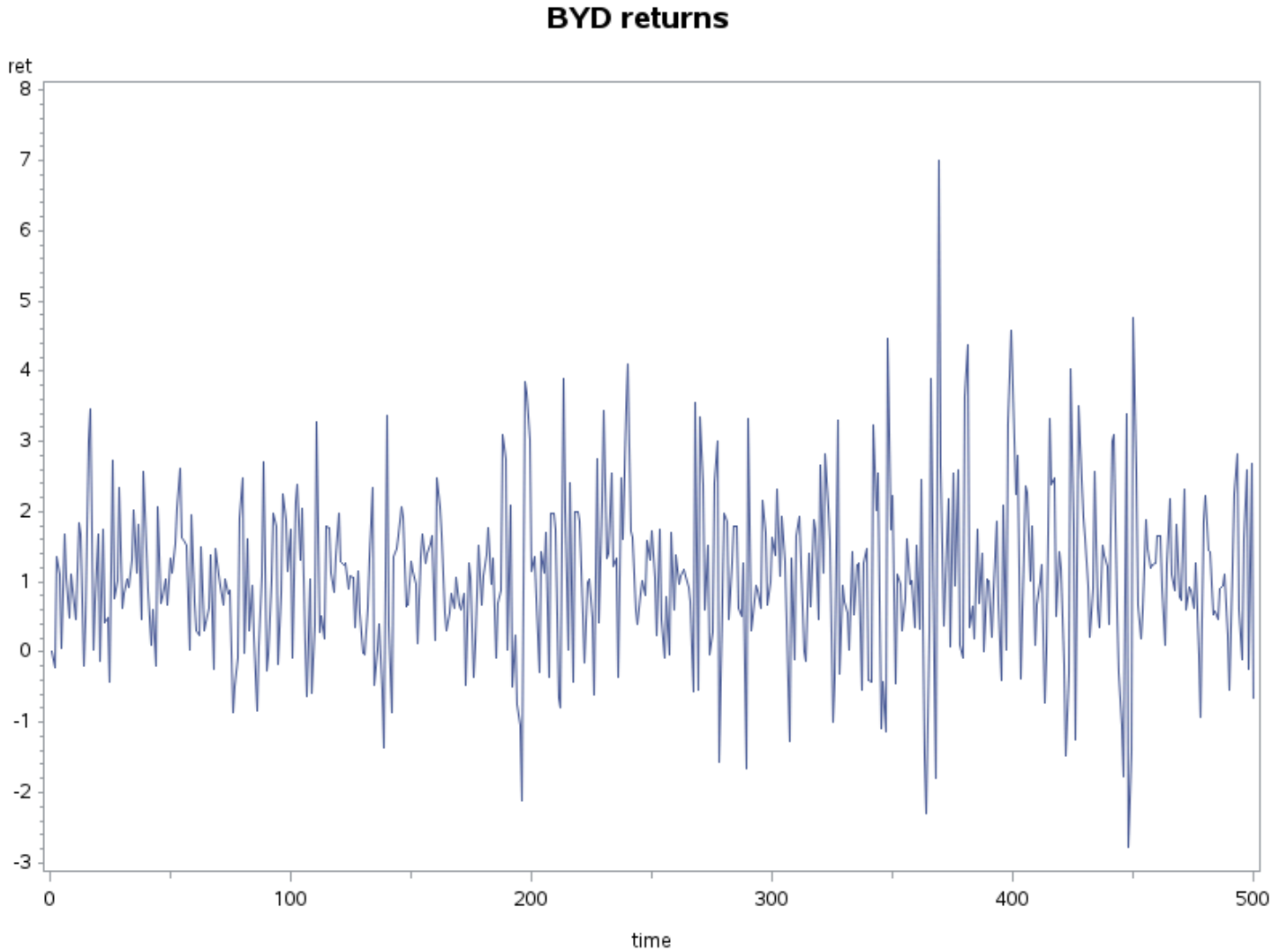
The UNIVARIATE Procedure



BYD returns

The UNIVARIATE Procedure


Q-Q Plot for ret
ret
8
6
4
2
0
-2
-4
-4
-3
-2
-1
0
1
2
3
4
Normal Quantiles



estimate mean of byd returns and save residuals

The AUTOREG Procedure

Dependent Variable	ret

estimate mean of byd returns and save residuals

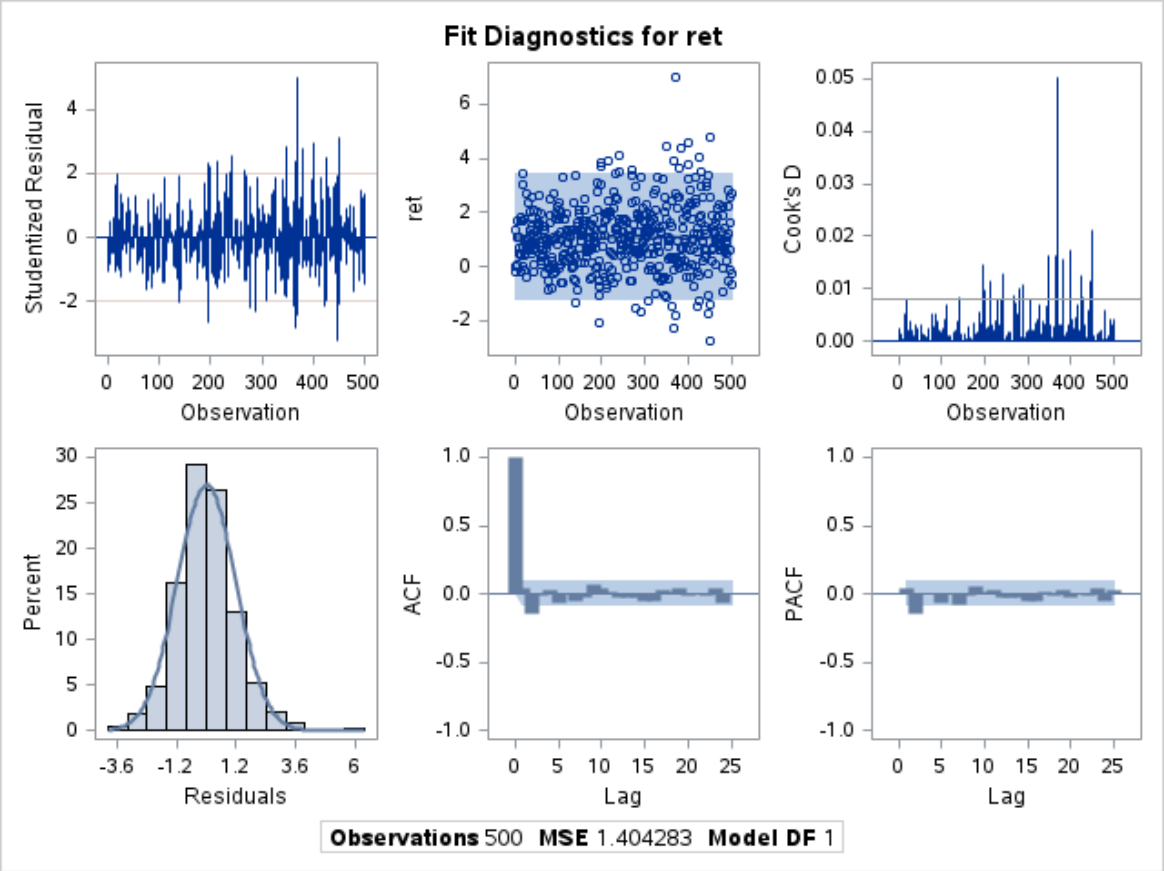
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	700.737278	**DFE**	499
MSE	1.40428	**Root MSE**	1.18502
SBC	1593.91561	**AIC**	1589.701
MAE	0.89611155	**AICC**	1589.70903
MAPE	472.633376	**HQC**	1591.35481
Durbin-Watson	1.9190	**Total R-Square**	0.0000

Parameter Estimates					
Variable	**DF**	**Estimate**	**Standard Error**	**t Value**	**Approx Pr > |t|**
Intercept	1	1.0783	0.0530	20.35	<.0001

estimate mean of byd returns and save residuals

The AUTOREG Procedure



test for ARCH effects in byd data

The AUTOREG Procedure

Dependent Variable	ehatsq

test for ARCH effects in byd data

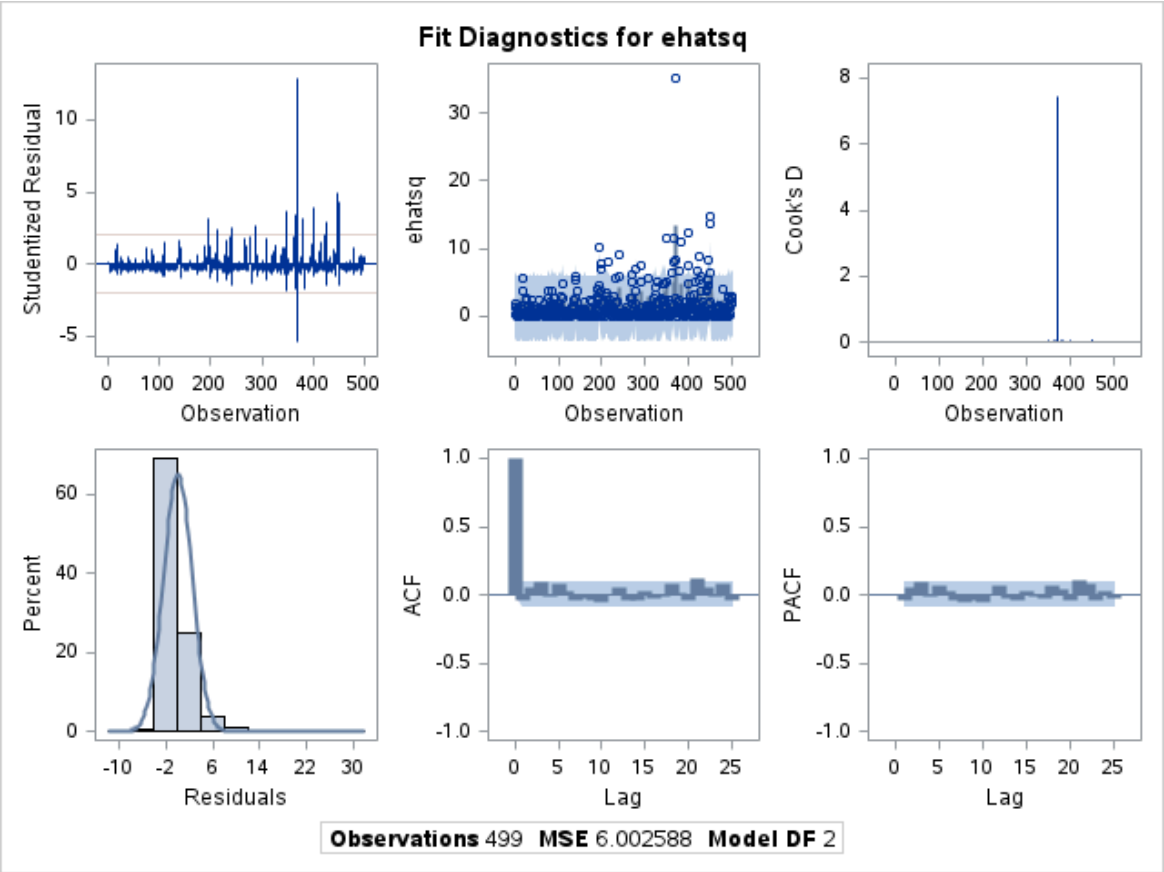
The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	2983.28626	**DFE**	497
MSE	6.00259	**Root MSE**	2.45002
SBC	2320.82502	**AIC**	2312.39981
MAE	1.37314066	**AICC**	2312.424
MAPE	2175618.42	**HQC**	2315.70613
Durbin-Watson	2.0670	**Total R-Square**	0.1246

Parameter Estimates					
Variable	**DF**	**Estimate**	**Standard Error**	**t Value**	**Approx Pr > |t|**
Intercept	1	0.9083	0.1244	7.30	<.0001
ehatsq1	1	0.3531	0.0420	8.41	<.0001

test for ARCH effects in byd data

The AUTOREG Procedure



estimate ARCH(1) model and fit volatility

The AUTOREG Procedure

Dependent Variable	ret

estimate ARCH(1) model and fit volatility

The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	700.737278	DFE	499
MSE	1.40428	Root MSE	1.18502
SBC	1593.91561	AIC	1589.701
MAE	0.89611155	AICC	1589.70903
MAPE	472.633376	HQC	1591.35481
Durbin-Watson	1.9190	Total R-Square	0.0000

Tests for ARCH Disturbances Based on OLS Residuals				
Order	Q	Pr > Q	LM	Pr > LM
1	62.6104	<.0001	62.2819	<.0001
2	84.4581	<.0001	66.2737	<.0001
3	97.7368	<.0001	68.5130	<.0001
4	100.3119	<.0001	68.7221	<.0001
5	105.0300	<.0001	70.4004	<.0001
6	106.0590	<.0001	70.5316	<.0001
7	106.4676	<.0001	72.2938	<.0001
8	107.6972	<.0001	73.0515	<.0001
9	109.6069	<.0001	73.3493	<.0001

Tests for ARCH Disturbances Based on OLS Residuals				
Order	Q	Pr > Q	LM	Pr > LM
10	111.5931	<.0001	73.5252	<.0001
11	111.7335	<.0001	73.9982	<.0001
12	111.9903	<.0001	75.1742	<.0001

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.0783	0.0530	20.35	<.0001

Algorithm converged.

estimate ARCH(1) model and fit volatility

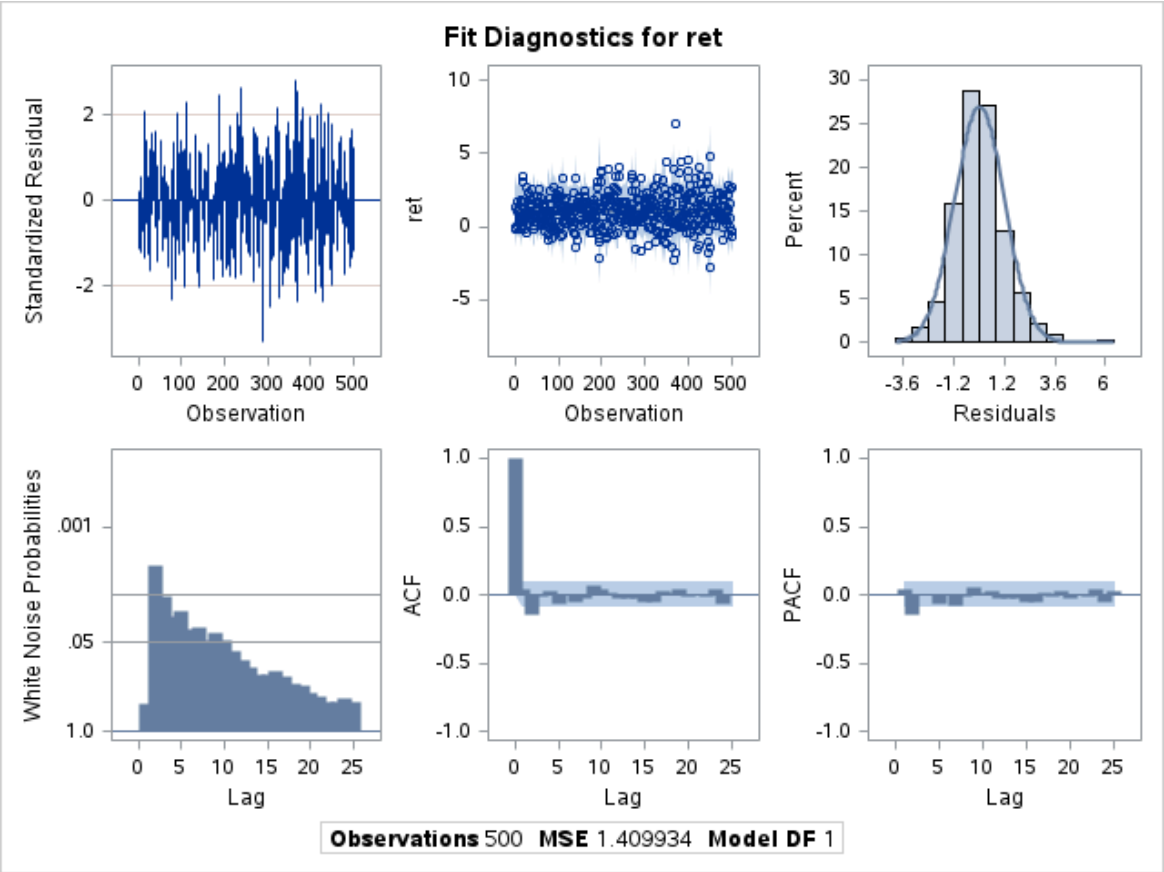
The AUTOREG Procedure

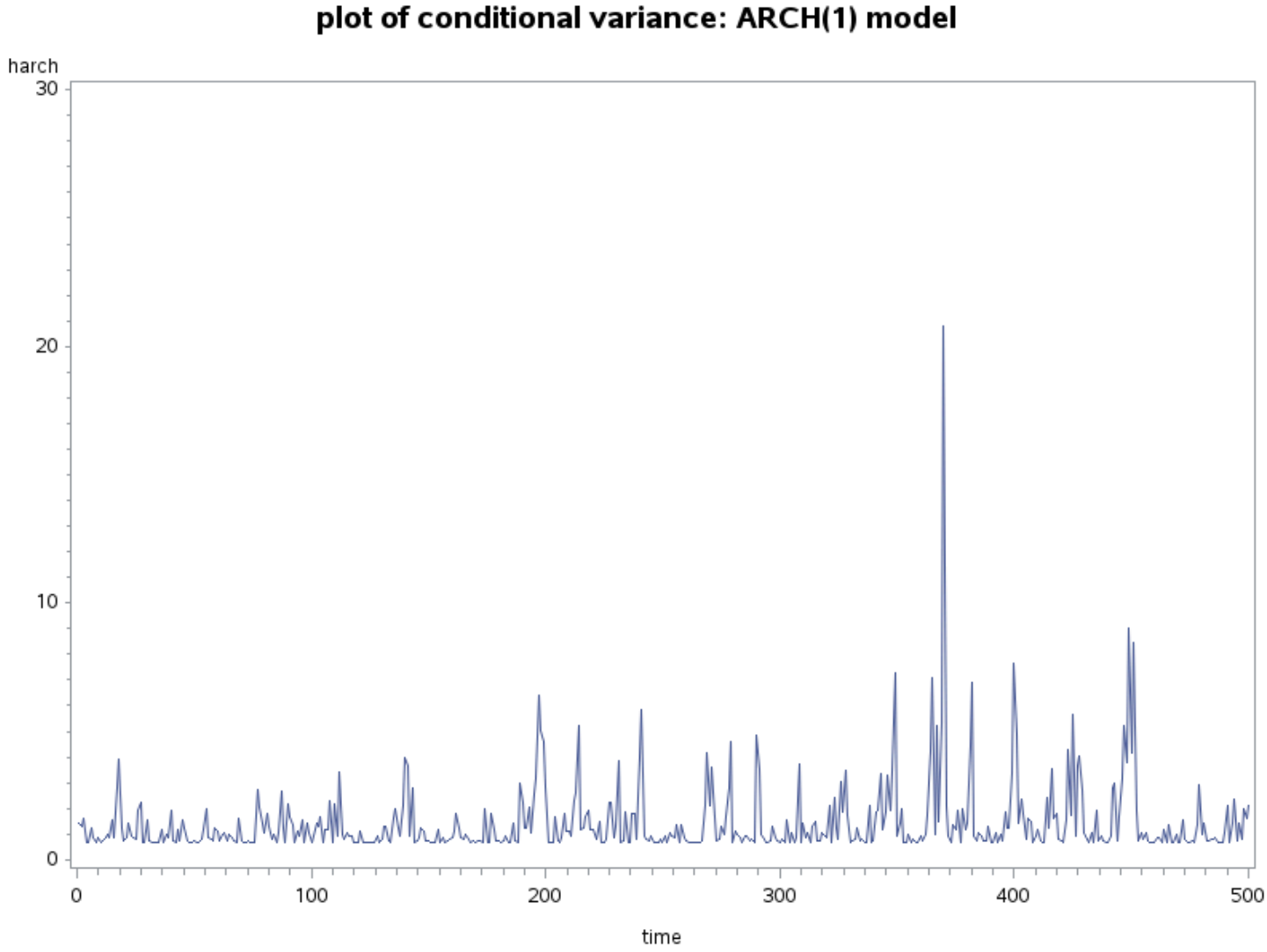
GARCH Estimates			
SSE	700.840315	Observations	500
MSE	1.40168	Uncond Var	1.49106919
Log Likelihood	-740.79331	Total R-Square	.
SBC	1500.23044	AIC	1487.58661
MAE	0.89555713	AICC	1487.635
MAPE	466.72465	HQC	1492.54803
		Normality Test	0.2817
		Pr > ChiSq	0.8686

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.0639	0.0396	26.89	<.0001
ARCH0	1	0.6421	0.0634	10.13	<.0001
ARCH1	1	0.5693	0.1032	5.52	<.0001

estimate ARCH(1) model and fit volatility

The AUTOREG Procedure


Fit Diagnostics for ret
Standardized Residual
Observation
ret
Observation
Percent
Residuals
White Noise Probabilities
Lag
ACF
Lag
PACF
Lag
Observations 500 MSE 1.409934 Model DF 1



estimate GARCH(1,1) model and fit volatility

The AUTOREG Procedure

Dependent Variable	ret

estimate GARCH(1,1) model and fit volatility

The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	700.737278	**DFE**	499
MSE	1.40428	**Root MSE**	1.18502
SBC	1593.91561	**AIC**	1589.701
MAE	0.89611155	**AICC**	1589.70903
MAPE	472.633376	**HQC**	1591.35481
Durbin-Watson	1.9190	**Total R-Square**	0.0000

Tests for ARCH Disturbances Based on OLS Residuals				
Order	**Q**	**Pr > Q**	**LM**	**Pr > LM**
1	62.6104	<.0001	62.2819	<.0001
2	84.4581	<.0001	66.2737	<.0001
3	97.7368	<.0001	68.5130	<.0001
4	100.3119	<.0001	68.7221	<.0001

Tests for ARCH Disturbances Based on OLS Residuals				
Order	Q	Pr > Q	LM	Pr > LM
5	105.0300	<.0001	70.4004	<.0001
6	106.0590	<.0001	70.5316	<.0001
7	106.4676	<.0001	72.2938	<.0001
8	107.6972	<.0001	73.0515	<.0001
9	109.6069	<.0001	73.3493	<.0001
10	111.5931	<.0001	73.5252	<.0001
11	111.7335	<.0001	73.9982	<.0001
12	111.9903	<.0001	75.1742	<.0001

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.0783	0.0530	20.35	<.0001

Algorithm converged.

estimate GARCH(1,1) model and fit volatility

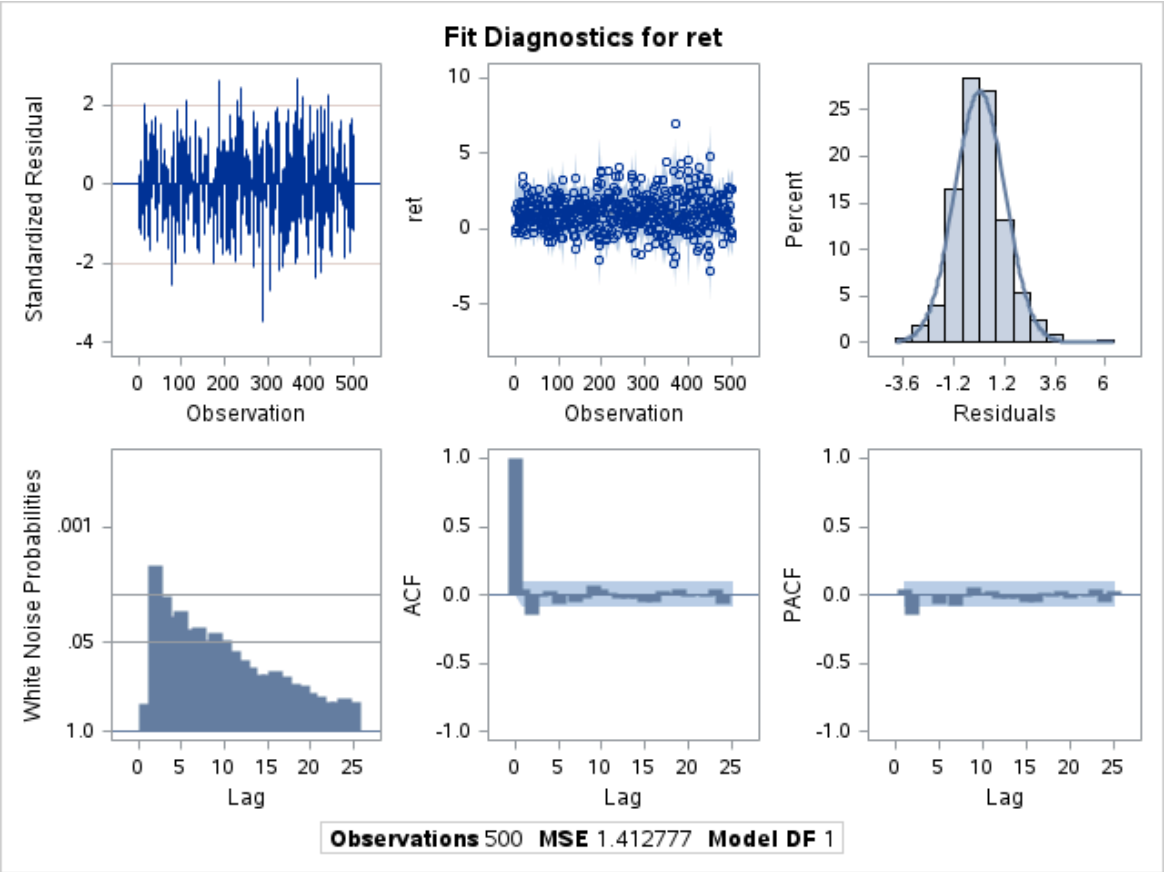
The AUTOREG Procedure

GARCH Estimates			
SSE	701.141027	Observations	500
MSE	1.40228	Uncond Var	1.47930255
Log Likelihood	-736.02876	Total R-Square	.
SBC	1496.91594	AIC	1480.05751
MAE	0.89519324	AICC	1480.13832
MAPE	460.953781	HQC	1486.67273
		Normality Test	0.9803
		Pr > ChiSq	0.6125

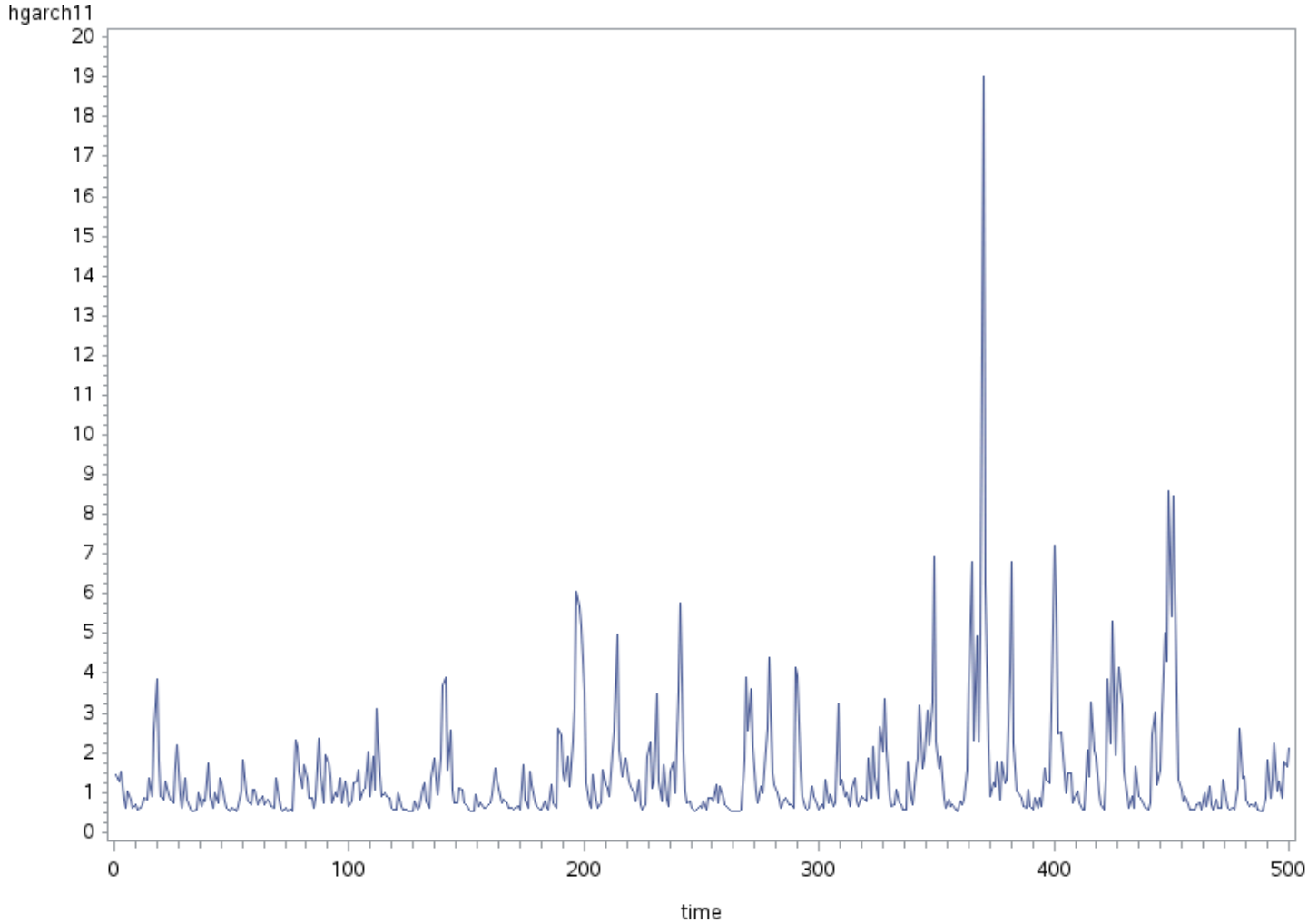
Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.0499	0.0406	25.84	<.0001
ARCH0	1	0.4011	0.0903	4.44	<.0001
ARCH1	1	0.4909	0.1019	4.82	<.0001
GARCH1	1	0.2380	0.1119	2.13	0.0335

estimate GARCH(1,1) model and fit volatility

The AUTOREG Procedure


Fit Diagnostics for ret
Standardized Residual
Observation
ret
Observation
Percent
Residuals
White Noise Probabilities
Lag
ACF
Lag
PACF
Lag
Observations 500 MSE 1.412777 Model DF 1

plot of conditional variance: GARCH(1,1) model



byd data with bad news indicator

Obs	ehat_garch	hgarch11	ehat_arch	harch	ehat	ret	time	ehatsq	ehatsq1	dt	dt1	ehat_gsq	ehat_gsq1	desq1
1	-1.04988	1.43097	-1.06394	1.44166	-1.07829	0.00000	1	1.16272	.	1	.	1.16272	.	.
2	-1.27707	1.28270	-1.29113	1.28662	-1.30549	-0.22719	2	1.70430	1.16272	1	1	1.70430	1.16272	1.16272
3	0.30097	1.50694	0.28690	1.59125	0.27255	1.35084	3	0.07428	1.70430	0	1	0.07428	1.70430	1.70430
4	0.05568	0.80416	0.04162	0.68900	0.02726	1.10556	4	0.00074	0.07428	0	0	0.00074	0.07428	0.00000
5	-0.98438	0.59397	-0.99844	0.64313	-1.01280	0.06550	5	1.02575	0.00074	1	0	1.02575	0.00074	0.00000

estimate T-GARCH(1,1) model and fit volatility

The AUTOREG Procedure

Dependent Variable	ret

estimate T-GARCH(1,1) model and fit volatility

The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	700.737278	DFE	499
MSE	1.40428	Root MSE	1.18502
SBC	1593.91561	AIC	1589.701

Ordinary Least Squares Estimates			
MAE	0.89611155	AICC	1589.70903
MAPE	472.633376	HQC	1591.35481
Durbin-Watson	1.9190	Total R-Square	0.0000

Tests for ARCH Disturbances Based on OLS Residuals				
Order	Q	Pr > Q	LM	Pr > LM
1	62.6104	<.0001	62.2819	<.0001
2	84.4581	<.0001	66.2737	<.0001
3	97.7368	<.0001	68.5130	<.0001
4	100.3119	<.0001	68.7221	<.0001
5	105.0300	<.0001	70.4004	<.0001
6	106.0590	<.0001	70.5316	<.0001
7	106.4676	<.0001	72.2938	<.0001
8	107.6972	<.0001	73.0515	<.0001
9	109.6069	<.0001	73.3493	<.0001
10	111.5931	<.0001	73.5252	<.0001
11	111.7335	<.0001	73.9982	<.0001
12	111.9903	<.0001	75.1742	<.0001

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.0783	0.0530	20.35	<.0001

Algorithm converged.

estimate T-GARCH(1,1) model and fit volatility

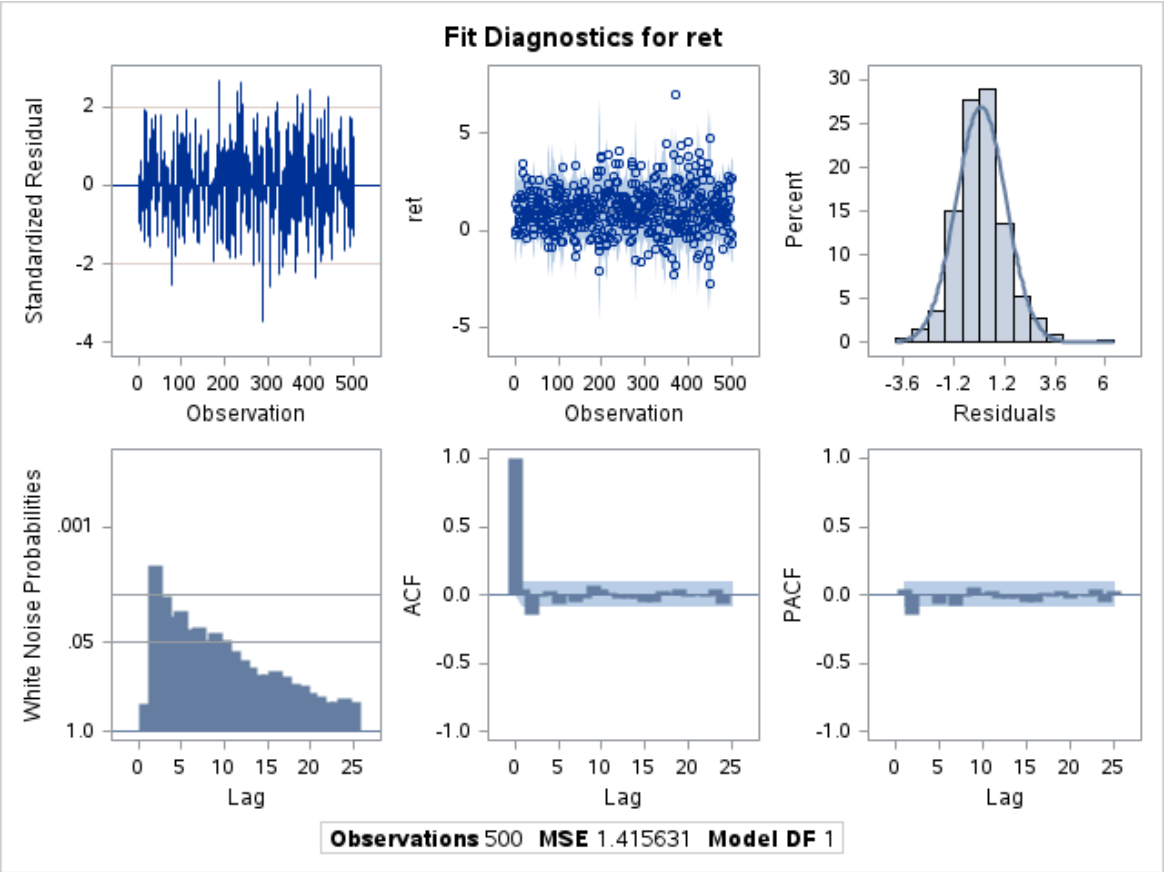
The AUTOREG Procedure

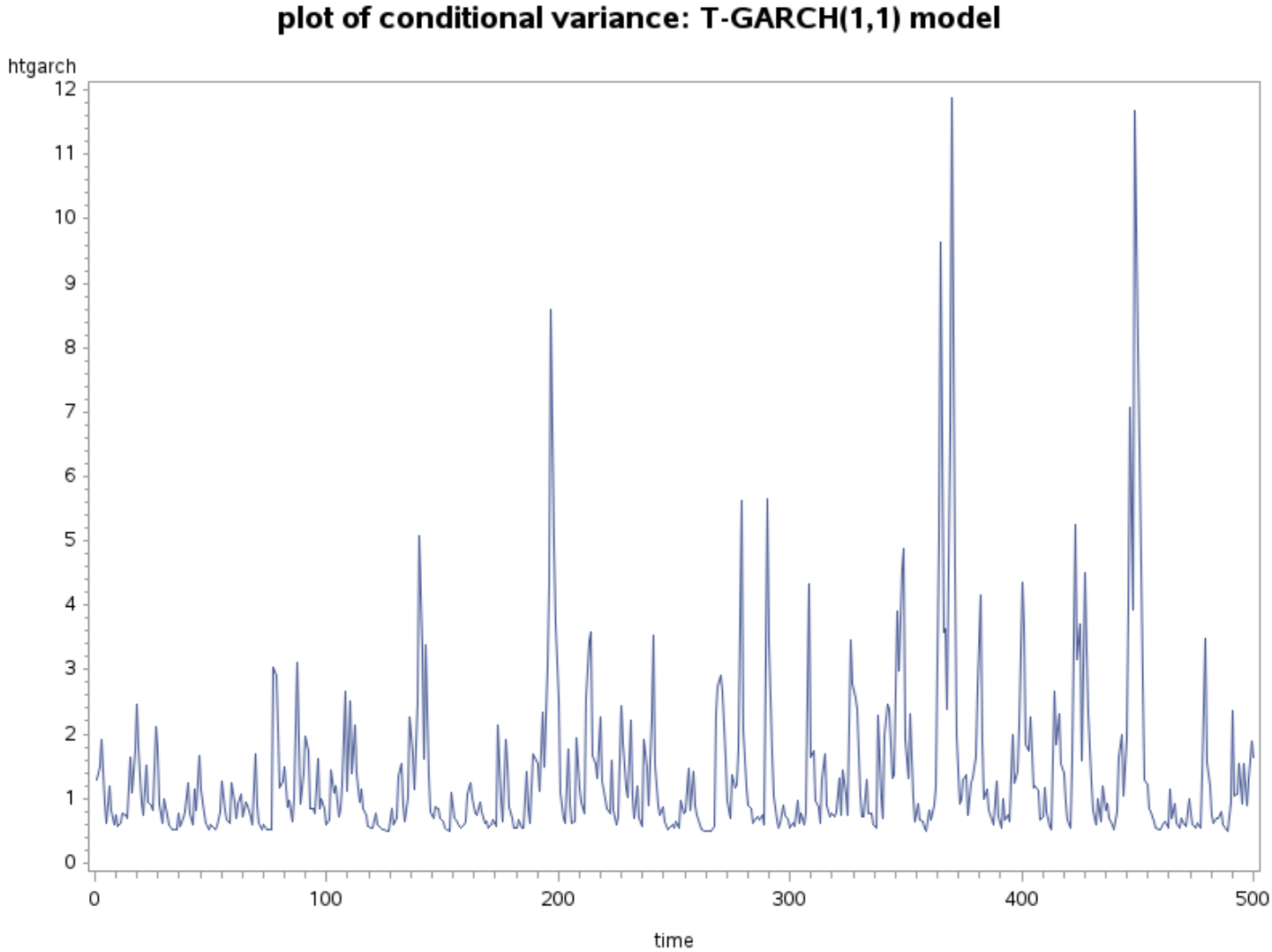
GARCH Estimates			
SSE	704.243336	Observations	500
MSE	1.40849	Uncond Var	0.77519654
Log Likelihood	-730.61004	Total R-Square	.
SBC	1492.29313	AIC	1471.22009
MAE	0.89560078	AICC	1471.34154
MAPE	438.427273	HQC	1479.48911
		Normality Test	0.5172
		Pr > ChiSq	0.7721

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	0.9946	0.0443	22.44	<.0001
ARCH0	1	0.3437	0.0924	3.72	0.0002
ARCH1	1	0.2651	0.0807	3.28	0.0010
GARCH1	1	0.2915	0.1165	2.50	0.0124
HET1	1	0.4348	0.1703	2.55	0.0107

estimate T-GARCH(1,1) model and fit volatility

The AUTOREG Procedure


Fit Diagnostics for ret
Standardized Residual
Observation
ret
Observation
Percent
Residuals
White Noise Probabilities
Lag
ACF
Lag
PACF
Lag
Observations 500 MSE 1.415631 Model DF 1



GARCH-M model

The AUTOREG Procedure

Dependent Variable	ret

GARCH-M model

The AUTOREG Procedure

Ordinary Least Squares Estimates			
SSE	700.737278	**DFE**	499
MSE	1.40428	**Root MSE**	1.18502
SBC	1593.91561	**AIC**	1589.701
MAE	0.89611155	**AICC**	1589.70903
MAPE	472.633376	**HQC**	1591.35481
Durbin-Watson	1.9190	**Total R-Square**	0.0000

Tests for ARCH Disturbances Based on OLS Residuals				
Order	**Q**	**Pr > Q**	**LM**	**Pr > LM**
1	62.6104	<.0001	62.2819	<.0001
2	84.4581	<.0001	66.2737	<.0001
3	97.7368	<.0001	68.5130	<.0001
4	100.3119	<.0001	68.7221	<.0001

Tests for ARCH Disturbances Based on OLS Residuals				
Order	Q	Pr > Q	LM	Pr > LM
5	105.0300	<.0001	70.4004	<.0001
6	106.0590	<.0001	70.5316	<.0001
7	106.4676	<.0001	72.2938	<.0001
8	107.6972	<.0001	73.0515	<.0001
9	109.6069	<.0001	73.3493	<.0001
10	111.5931	<.0001	73.5252	<.0001
11	111.7335	<.0001	73.9982	<.0001
12	111.9903	<.0001	75.1742	<.0001

Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	1.0783	0.0530	20.35	<.0001

Algorithm converged.

GARCH-M model

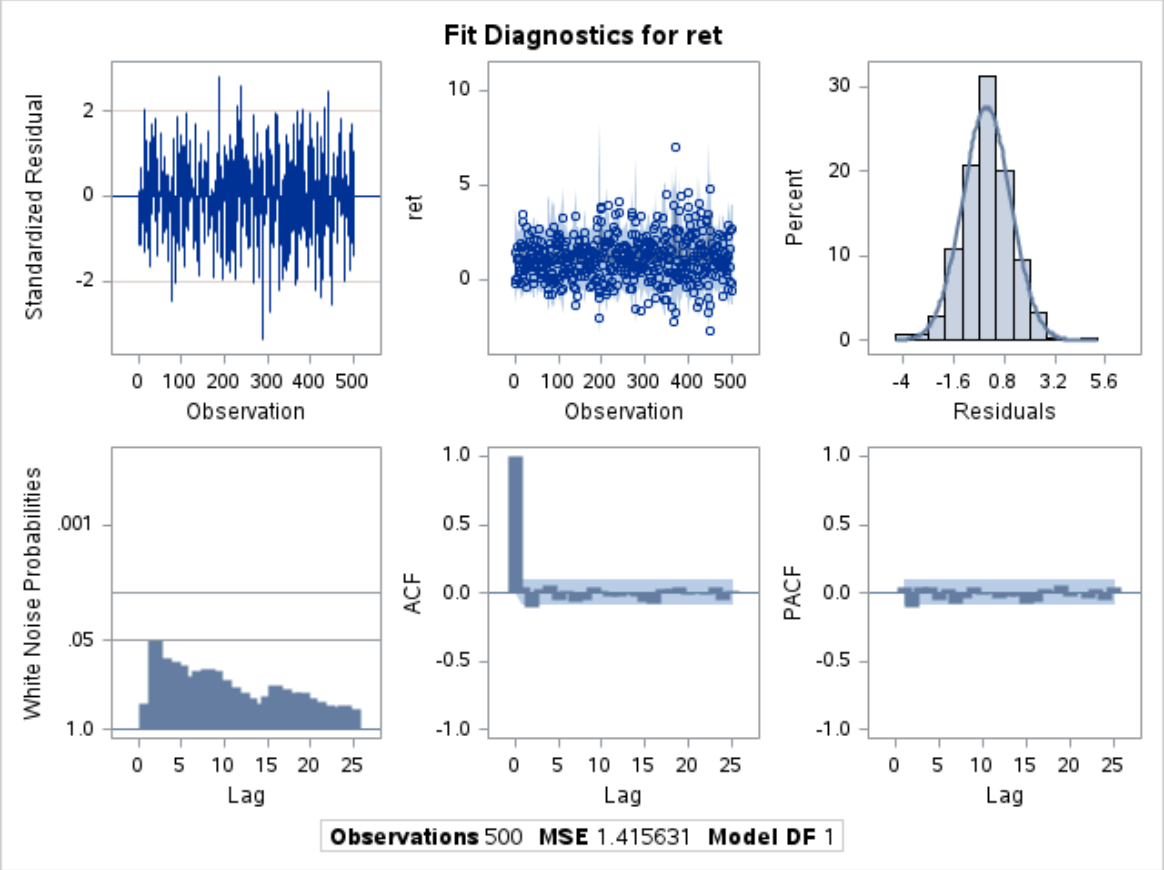
The AUTOREG Procedure

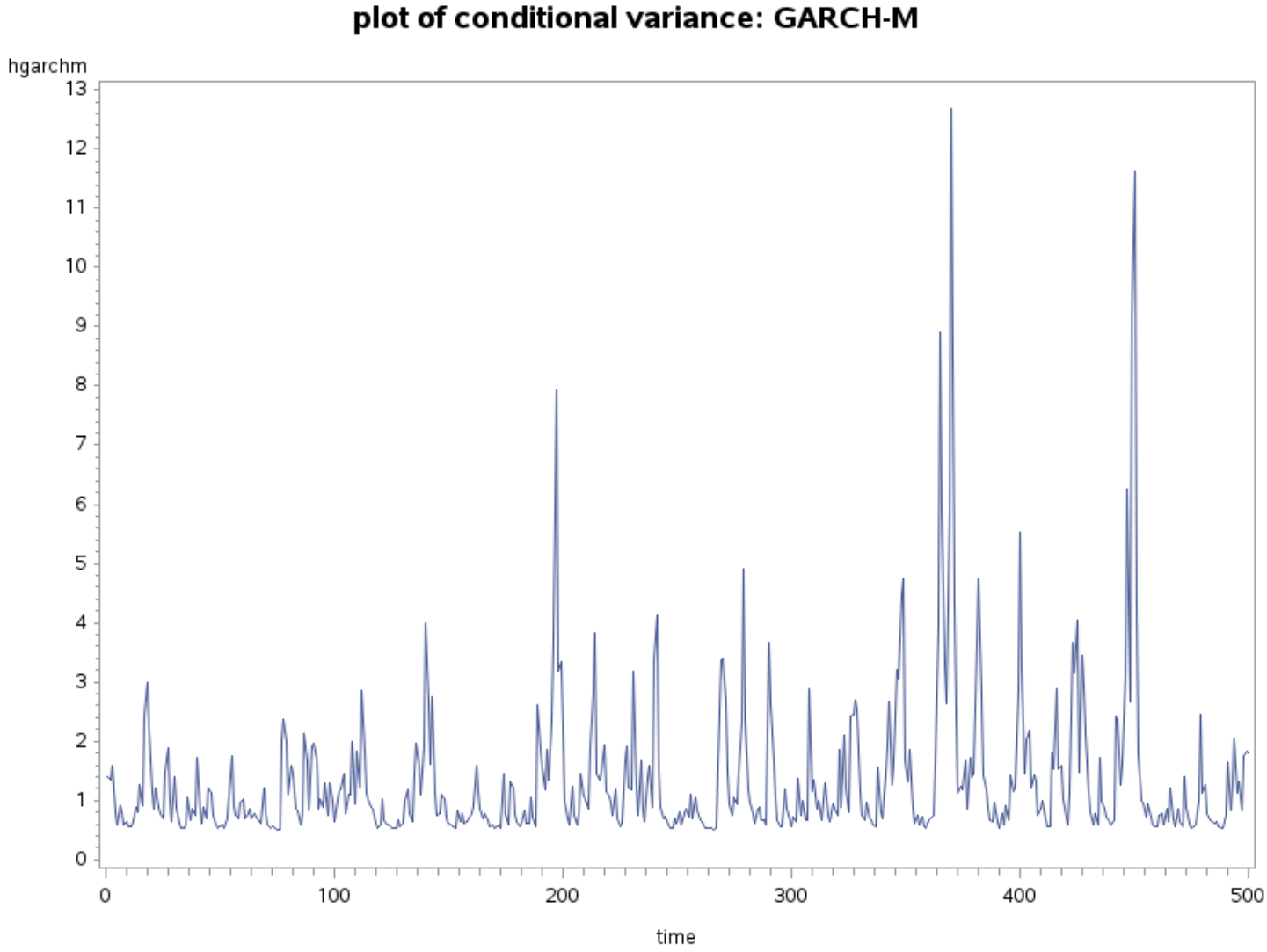
GARCH Estimates			
SSE	671.611966	Observations	500
MSE	1.34322	Uncond Var	.
Log Likelihood	-727.71862	Total R-Square	0.0416
SBC	1486.51028	AIC	1465.43724
MAE	0.88298581	AICC	1465.5587
MAPE	485.825117	HQC	1473.70627
		Normality Test	4.4117
		Pr > ChiSq	0.1102

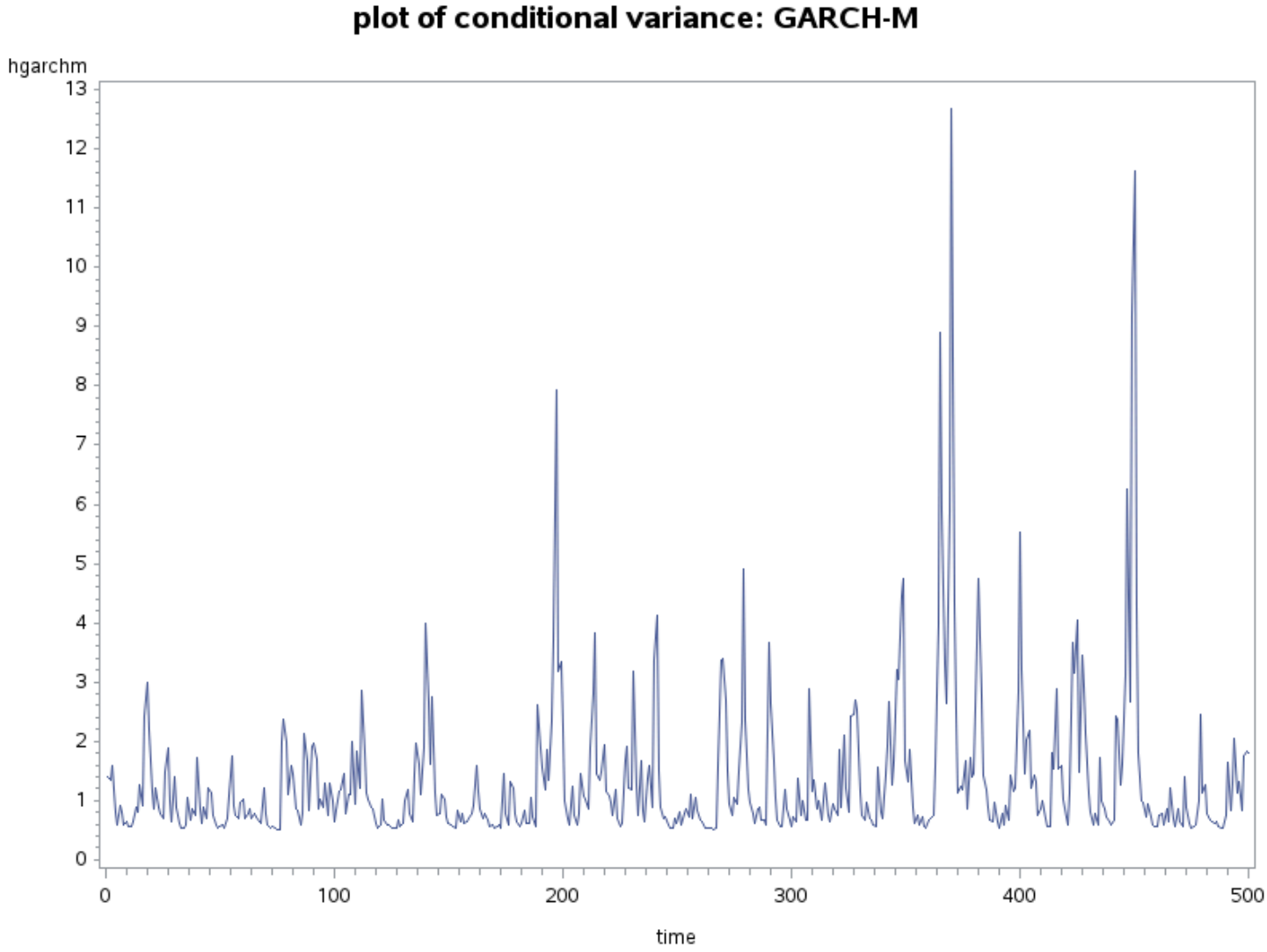
Parameter Estimates					
Variable	DF	Estimate	Standard Error	t Value	Approx Pr > |t|
Intercept	1	0.8212	0.0729	11.26	<.0001
ARCH0	1	0.3726	0.0858	4.34	<.0001
ARCH1	1	0.4584	0.1028	4.46	<.0001
GARCH1	1	0.2717	0.1107	2.45	0.0141
DELTA	1	0.2309	0.0664	3.48	0.0005

GARCH-M model

The AUTOREG Procedure


Fit Diagnostics for ret
Standardized Residual
Observation
ret
Observation
Percent
Residuals
White Noise Probabilities
Lag
ACF
Lag
PACF
Lag
Observations 500 MSE 1.415631 Model DF 1


plot of conditional variance: GARCH-M
hgarchm
13
12
11
10
9
8
7
6
5
4
3
2
1
0
0
100
200
300
400
500
time


plot of conditional variance: GARCH-M
hgarchm
13
12
11
10
9
8
7
6
5
4
3
2
1
0
0
100
200
300
400
500
time